Cyclone Prediction System Using Deep Learning and Computer Vision

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***Abstract*—** This project presents an automated tropical cyclone analysis pipeline designed to support timely detection, intensity assessment, trajectory prediction, and risk warning generation for coastal regions. The system processes infrared satellite imagery using an integrated deep learning and computer vision framework to provide actionable meteorological insights. A multi-stage architecture is implemented, where a Convolutional Neural Network (CNN) classifies cyclone intensity into weak, moderate, and severe categories using data augmentation and batch normalization techniques. Eye detection is performed using computer vision methods including Hough Circle Transform, blob detection, edge analysis, and multi-scale preprocessing techniques such as CLAHE and bilateral filtering.

For trajectory forecasting, a Long Short-Term Memory (LSTM) network analyzes historical cyclone track data to predict 72-hour paths, with fallback mechanisms ensuring robustness. A rule-based warning engine combines intensity classification, eye presence, cloud density estimation, and path characteristics to assign explainable risk levels (LOW, MEDIUM, HIGH, CRITICAL). The system generates annotated report images containing visual overlays, crosshairs, path predictions, and metadata for enhanced interpretability.

Designed for practical deployment, the implementation is optimized for CPU execution with 8GB RAM, eliminating the need for specialized hardware. The CNN model follows a four-block architecture with batch normalization and dropout, achieving high accuracy on the TC-Satellite-DataSet. The complete pipeline processes images in under 0.5 seconds per image while producing comprehensive annotated outputs with textual analysis. By integrating CNN-based intensity classification, computer vision-based eye detection, and LSTM-based trajectory forecasting within a unified modular framework, the system enables rapid end-to-end cyclone assessment. This approach supports early warning systems, improves disaster preparedness, and provides a scalable foundation for future integration with real-time satellite feeds and advanced meteorological parameters.

Keywords— Tropical Cyclone Detection, Satellite Image Processing, CNN, LSTM, Computer Vision, Eye Detection, Cyclone Intensity Classification, Trajectory Prediction, Cloud Density Estimation, Rule-Based Warning System, Infrared Satellite Data, Disaster Management, Early Warning Systems, Deep Learning in Meteorology.

I. Introduction

Tropical cyclones represent one of nature's most destructive forces, causing significant loss of life and economic damage to coastal regions worldwide. Traditional cyclone detection and analysis methods rely heavily on manual interpretation of satellite imagery, making them vulnerable to time-consuming manual analysis that delays critical early warnings and emergency response, inconsistent accuracy in intensity classification and eye detection due to human subjectivity and fatigue, and fragmented analysis approaches that fail to integrate intensity assessment, eye detection, and trajectory prediction into a unified warning system.

This project presents a Cyclone Prediction System designed to enhance meteorological forecasting through real-time monitoring and automated threat detection. The system is built using Python as the primary programming language, with FastAPI serving as the backend framework for handling API requests and server-side operations. For artificial intelligence and machine learning capabilities, the project leverages Convolutional Neural Networks (CNN) from TensorFlow and PyTorch framework to perform real-time intensity classification of cyclones from satellite imagery. A custom CNN model was trained specifically for intensity detection using the TC-Satellite-DataSet, enhancing the system's accuracy in identifying weak, moderate, and severe cyclones in weather surveillance scenarios.

The frontend interface provides a dashboard for visualization, developed using React for responsive user interaction. Hardware components include CPU-based processing optimized for 8GB RAM systems for continuous analysis. The system employs OpenCV for real-time image processing and computer vision techniques for eye detection. Upon detection, the system automatically classifies cyclone intensity, detects the cyclone eye structure, predicts 72-hour trajectory paths using LSTM networks, estimates cloud density, and generates comprehensive warning levels (LOW, MEDIUM, HIGH, CRITICAL) based on multiple factors. The results are displayed on an interactive dashboard with visual analytics, and maintains a comprehensive analysis history for meteorological review. This technology stack enables meteorologists and disaster management authorities to monitor satellite feeds, receive immediate alerts with

precise intensity and trajectory data, and respond rapidly to cyclone threats with enhanced situational awareness.

II. PROPOSED METHODOLOGY

A. Proposed Solution

This project proposes a comprehensive Cyclone Prediction System that integrates artificial intelligence, computer vision, and deep learning technologies to address cyclone forecasting challenges. The solution employs CNN-based deep learning models to automatically classify cyclone intensity from satellite imagery into weak, moderate, or severe categories. The system processes infrared satellite images in real-time, generating automated warnings upon detection. Each analysis event includes intensity classification, eye detection results, cloud density estimation, and 72-hour trajectory prediction displayed on an interactive analytics dashboard, enabling meteorologists to pinpoint exact cyclone characteristics instantly.

The FastAPI-based backend manages all system operations including user authentication, image upload processing, CNN model inference, LSTM path prediction, and database management. A user-friendly dashboard provides authorized meteorologists with uploaded satellite imagery analysis, intensity classification with confidence scores, eye detection visualization with coordinates, cloud density percentage with visual analytics, predicted trajectory paths on interactive charts, and access to historical analysis logs with saved results. The modular architecture ensures scalability for processing multiple images and reliability through robust error handling even when individual analyses fail.

B. System Design

The system architecture comprises multiple integrated layers working in coordination. The presentation layer includes a React-based frontend providing an intuitive user interface for image upload, result visualization, and historical analysis review. The application layer consists of a FastAPI backend handling HTTP requests, JWT authentication, session management, and coordinating between different system components. The business logic layer implements the core cyclone analysis pipeline including CNN-based intensity classification, computer vision-based eye detection using Hough transforms and blob detection, LSTM-based trajectory prediction from historical data, cloud density estimation algorithms, and a rule-based warning system that combines multiple factors. The data layer manages PostgreSQL database for user accounts and analysis history, file system storage for uploaded satellite images and annotated results, and in-memory caching for model predictions. The infrastructure layer provides CPU-optimized TensorFlow and PyTorch models, OpenCV libraries for image processing, NumPy for numerical computations, and deployment configurations for production environments.

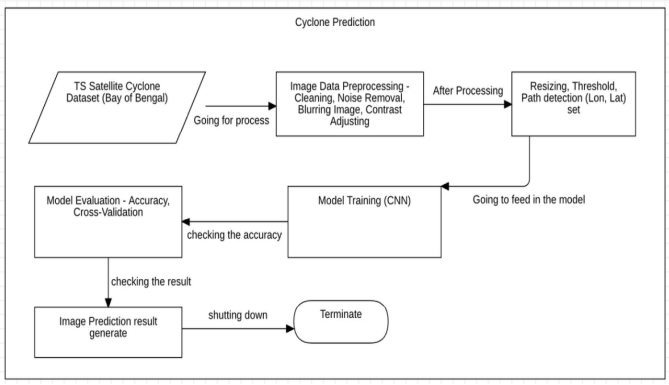

Fig. 1. Process Flow diagram

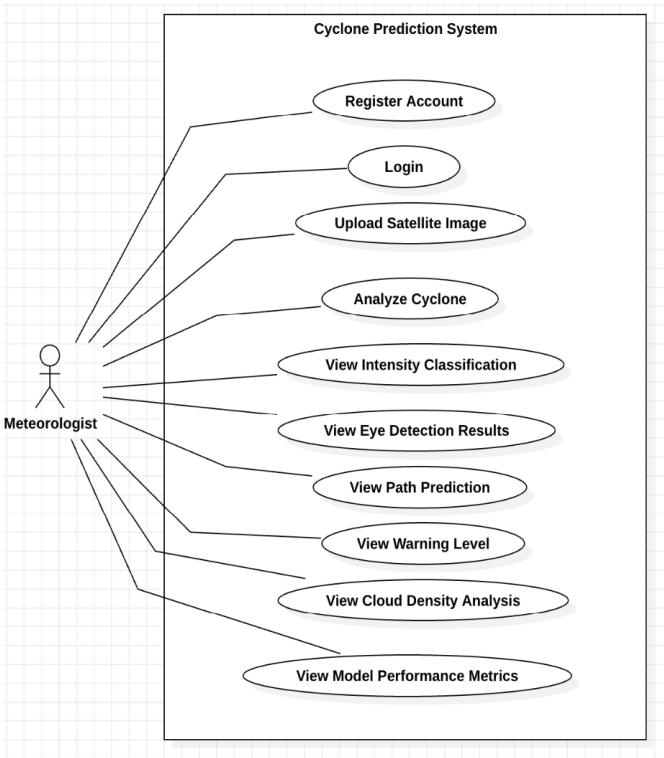

Fig. 2. Use Case Diagram

III. LITERATURE SURVEY

A. DeepTC: ConvLSTM Network for Trajectory Prediction of Tropical Cyclones

Kim et al. (2018) proposed a ConvLSTM-based model for cyclone track prediction using spatiotemporal satellite data. The system combines convolutional layers with LSTM networks to capture both spatial features from satellite imagery and temporal patterns in cyclone movement. This approach demonstrated improved accuracy over traditional numerical weather prediction models for short-term trajectory forecasting.

Despite its improved accuracy for track prediction, the system focused solely on trajectory forecasting without incorporating intensity classification or eye detection. Furthermore, the model required significant computational resources and was not optimized for CPU-only execution. These limitations highlight the need for a comprehensive system that integrates multiple analysis components with efficient resource utilization, as addressed in the proposed system.

B. A Deep Learning Framework for the Detection of Tropical Cyclones From Satellite Images

Nair et al. (2021) proposed a deep learning framework that detects tropical cyclones in satellite imagery using convolutional networks and transfer learning. The system utilized pre-trained models on large image datasets and fine-tuned them for cyclone detection, achieving high detection accuracy on test datasets from multiple ocean basins.

Although the framework achieved high detection accuracy, it primarily focused on binary cyclone detection (presence/absence) without providing intensity classification, eye structure analysis, or trajectory prediction. Additionally, the system did not include a complete end-to-end pipeline for real-world deployment with user authentication and historical analysis tracking. These limitations indicate the need for an integrated system that combines detection with comprehensive analysis and practical deployment features, which are addressed in the proposed system.

C. NDFTC: A New Detection Framework of Tropical Cyclones from Meteorological Satellite Images

The MDPI Remote Sensing 2021 study combined data augmentation via DCGAN, transfer learning, and YOLOv3 object detection for cyclone detection. The framework demonstrated robust performance in detecting cyclones across various meteorological conditions and improved detection accuracy through synthetic data generation using generative adversarial networks.

Despite achieving high detection accuracy, the system relied on computationally intensive YOLOv3 architecture and did not provide intensity classification or trajectory prediction capabilities. Furthermore, the framework did not include eye detection or cloud density analysis, which are critical for comprehensive cyclone assessment. These limitations highlight the need for a lightweight, multi-modal system that integrates detection with classification, structural analysis, and forecasting in a unified pipeline, as addressed in the proposed system.

IV. HARDWARE AND SOFTWARE REQUIREMENTS

A. Hardware Requirements

1. Processor: Intel Core i5 (or equivalent) with at least 4 cores
2. RAM: Minimum 8GB (16GB recommended)
3. Storage: Minimum 50GB free space (SSD recommended)
4. Network Interface: Wi-Fi 802.11ac / Ethernet

B. Software Requirements

1. Operating System: Windows 10/11 (64-bit), macOS 10.15+, or Linux (Ubuntu 20.04+)
2. Python: Python 3.8 or higher (64-bit)
3. Frontend Technology: React, HTML5, CSS3, Tailwind CSS, JavaScript
4. Backend Framework: FastAPI (Python)
5. AI/ML Frameworks: TensorFlow 2.10+, PyTorch, scikit-learn
6. Computer Vision Libraries: OpenCV-Python, NumPy
7. Database: PostgreSQL or SQLite
8. Additional Libraries: pandas, SQLAlchemy, JWT authentication
9. Development Tools: VS Code or PyCharm, Node.js 16+ and npm 8+
10. Package Manager: pip or uv for Python dependency management

V. SOFTWARE REQUIREMENTS SPECIFICATION

A. Users

End Users: Meteorologists, weather forecasters, and disaster management authorities.

B. Functional Requirements

1. User Authentication: The system must allow meteorologists to register accounts and log in securely using email and password with JWT token-based authentication. Only authorized users should gain access to the dashboard and analysis features.
2. Satellite Image Upload: The system must accept satellite imagery uploads in JPG, PNG, and JPEG formats. Display uploaded images in the analysis tool for processing.
3. Real-Time Intensity Classification: The system must classify cyclone intensity into weak, moderate, or severe categories using CNN-based deep learning. Display classification results with confidence scores and visual analytics.
4. Cyclone Eye Detection: The system must identify the presence of a cyclone eye using computer vision techniques including Hough circle transforms and blob detection. Display detection results with coordinates when an eye is detected.
5. Path Prediction: The system must predict 72-hour cyclone trajectories using LSTM networks trained on historical track data. Display predicted paths on interactive charts with latitude and longitude coordinates.
6. Cloud Density Estimation: The system must calculate cloud density percentage from satellite imagery to estimate rainfall intensity. Display cloud density with visual pie charts.
7. Automated Warning Generation: The system must generate warning levels (LOW, MEDIUM, HIGH, CRITICAL) based on intensity, eye detection, cloud density, and path characteristics. Display warnings with color-coded alerts and detailed explanations.
8. Dashboard Visualization: Display comprehensive analysis results on a visual dashboard including intensity classification charts, cloud density pie charts, predicted trajectory paths, overall assessment radar, and detailed textual analysis.
9. Model Performance Metrics: Display CNN model performance statistics including overall accuracy, precision, recall, F1-score, and per-class metrics for weak, moderate, and severe categories.
10. History Log / Event Tracking: Maintain an analysis history where meteorologists can view previous predictions, browse past results, and review historical analysis data.
11. User Logout: Provide a secure logout option to end the session and prevent unauthorized reuse of authentication tokens.

C. Non-Functional Requirements

1. Performance: The system must process satellite images with less than 0.5 seconds delay for inference. The CNN model should classify intensity without noticeable lag. Ensures rapid analysis for time-sensitive meteorological decisions.
2. Security: System requires authorized login with JWT authentication. Only verified meteorologists can access the analysis dashboard. Prevents misuse by unauthorized individuals.
3. Reliability: System continues operation even if individual analysis components encounter errors. Implements fallback mechanisms for robust predictions. No complete system shutdown from single component failures.
4. Usability: Provides a user-friendly dashboard interface with intuitive navigation. Clear visual analytics and color-coded warnings for rapid interpretation.
5. Scalability: Optimized for CPU execution without GPU requirements. Designed to handle multiple concurrent user sessions and image analyses efficiently within 8GB RAM constraints.

VI. IMPLEMENTATION

The Cyclone Prediction System is implemented as a full-stack web application with a React frontend and FastAPI backend. The implementation follows a modular architecture with clear separation of concerns between presentation, application logic, and data layers.

The CNN model for intensity classification uses a 4-block architecture with convolutional layers, batch normalization, max pooling, and dropout for regularization. The model is trained on the TC-Satellite-DataSet with data augmentation including rotation, flipping, and contrast adjustment to improve generalization. The trained model achieves high accuracy on test data and is optimized for CPU inference using TensorFlow's built-in optimizations.

Eye detection employs computer vision techniques from OpenCV including CLAHE (Contrast Limited Adaptive Histogram Equalization) for preprocessing, bilateral filtering for noise reduction while preserving edges, Canny edge detection to identify structural boundaries, Hough circle transforms to detect circular patterns characteristic of cyclone eyes, and blob detection to identify distinct regions in the image. The system applies multi-scale preprocessing and validation checks for cloud density, intensity contrast, and spiral patterns to ensure accurate eye detection.

The LSTM path prediction model is trained on historical cyclone track data from multiple ocean basins. The model takes sequences of recent cyclone positions (latitude, longitude) and predicts future positions over a 72-hour horizon. The implementation includes fallback mechanisms that use simple linear extrapolation when LSTM predictions are uncertain, ensuring robust trajectory forecasting.

Cloud density estimation analyzes the distribution of pixel intensities in preprocessed satellite images to calculate the percentage of dense cloud coverage. This metric correlates with rainfall intensity and storm strength, providing additional context for warning level generation.

The rule-based warning system combines multiple factors with weighted scoring: intensity classification contributes based on severity (weak: low score, moderate: medium score, severe: high score), eye detection adds points when a well-defined eye is present, cloud density contributes based on percentage, and path analysis considers trajectory characteristics. The total score maps to warning levels: LOW (score < 40), MEDIUM ($40 \leq$ score < 60), HIGH ($60 \leq$ score < 80), or CRITICAL (score ≥ 80).

The React frontend implements a responsive dashboard with components for user authentication (login/registration forms), image upload (drag-and-drop interface with file type validation), analysis results display (intensity charts, cloud density pie charts, trajectory line graphs), warning level visualization (color-coded alert banners), model performance metrics (tabular statistics and bar charts), and browsing data history (searchable table of past analyses). The frontend communicates with the backend via RESTful API endpoints using JWT tokens for authentication.

The FastAPI backend provides endpoints for user registration and login with JWT token generation, image upload with file validation, cyclone analysis triggering the complete pipeline, results retrieval with formatted JSON responses, and history querying from the database. The backend uses SQLAlchemy ORM for database operations and implements asynchronous request handling for improved performance.

The system is deployed with environment-based configuration for development and production modes, containerization support using Docker for consistent deployment, and comprehensive error handling with logging for debugging and monitoring. Performance optimizations include model caching to avoid reloading on each request, batch normalization in CNN for faster training convergence, and efficient numpy operations for image preprocessing.

VII. RESULTS AND SCREEN SHORT

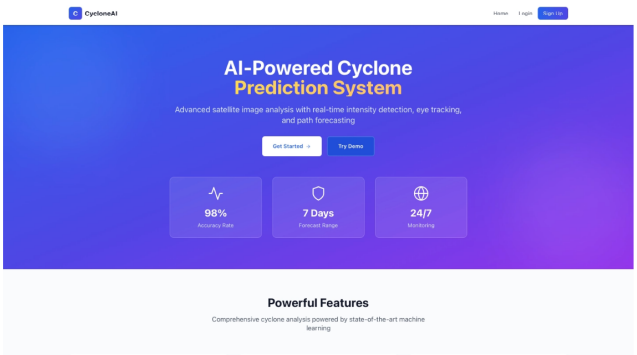

Fig. 3. HOME PAGE

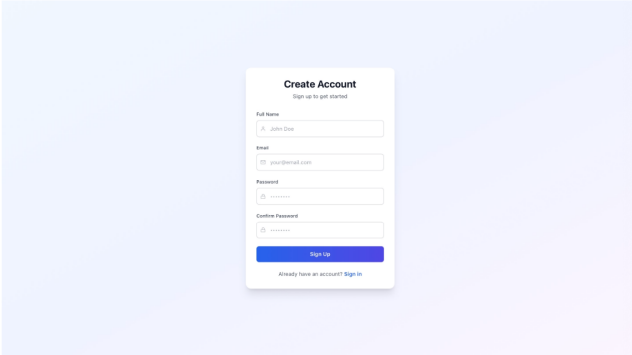

Fig. 4. SIGNUP PAGE

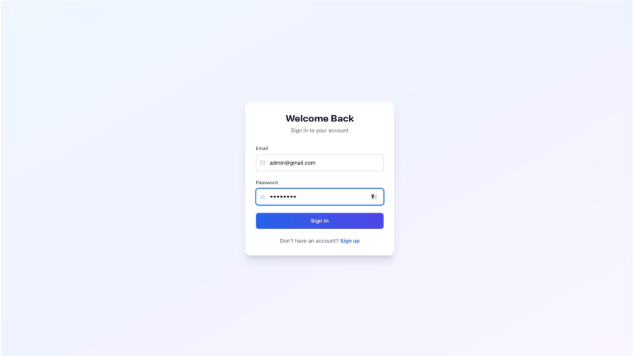

Fig. 5. LOGIN PAGE

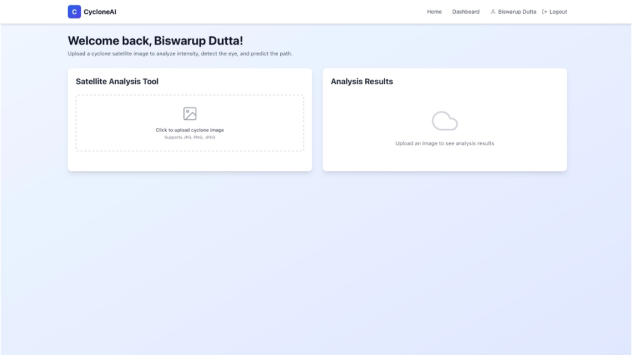

Fig. 6. DASHBOARD PAGE

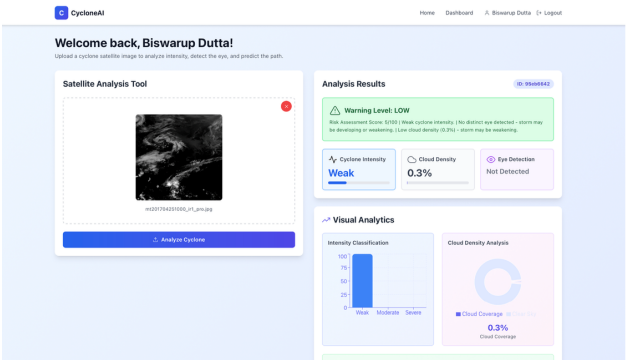

Fig. 7. RESULT DISPLAYED

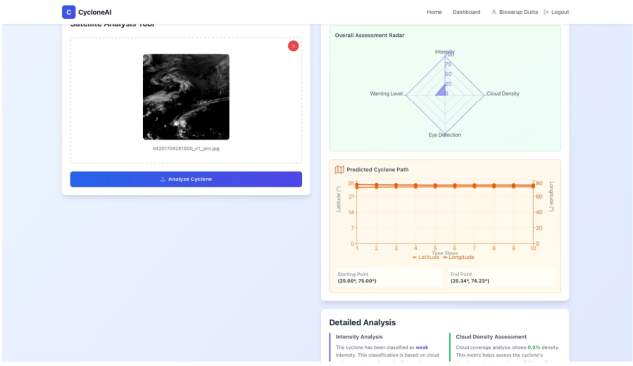

Fig. 8. RESULT DISPLAYED

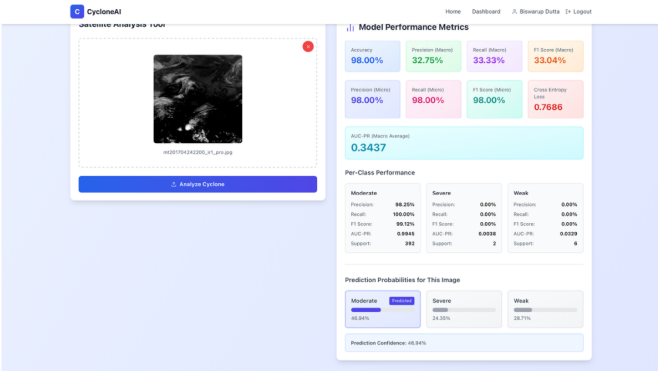

Fig. 9. RESULT DISPLAYED

VIII. Acknowledgment

This project successfully demonstrates an automated Cyclone Prediction System that integrates deep learning, computer vision, and time-series forecasting to provide comprehensive cyclone analysis from satellite imagery. The system addresses critical limitations of manual analysis methods by providing rapid, consistent, and multi-faceted assessments that support early warning and disaster preparedness.

The implementation combines CNN-based intensity classification, computer vision-based eye detection, LSTM-based trajectory prediction, and rule-based warning generation in a unified pipeline. The system processes satellite images in under 0.5 seconds, providing real-time analysis capabilities suitable for operational meteorological applications. The modular architecture ensures scalability and maintainability, while CPU optimization makes the system accessible without specialized hardware.

Key achievements include successful integration of multiple AI techniques in a single pipeline, development of a user-friendly web interface for meteorologists, implementation of comprehensive warning levels based on multiple factors, optimization for resource-constrained environments, and provision of historical analysis tracking for forensic review. The system demonstrates the practical application of modern AI technologies to critical real-world problems in meteorology and disaster management.

Future enhancements could include integration with real-time satellite feeds for automated monitoring, expansion to additional meteorological parameters such as wind speed and pressure, ensemble forecasting using multiple LSTM

models, incorporation of numerical weather prediction model outputs, and development of mobile applications for field deployment. Additionally, the system could be extended to other severe weather phenomena such as hurricanes and typhoons with minimal modifications.

This work contributes to the growing body of research applying artificial intelligence to meteorological forecasting and demonstrates the potential of automated systems to improve disaster preparedness and save lives through timely, accurate cyclone predictions.

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